H3 – First Steps in Your Cyber Lab

x) Read/watch/listen and summarize

Heinonen 2026: How to Install Linux in VirtualBox?

- The guide walks through installing Debian Linux inside VirtualBox on a Windows host.
- **Step one:** download the Debian live ISO. The example in the guide is debian-live-12.6.0-amd64-xfce.iso, but the link changes with new versions; always use the current Live Xfce image from cdimage.debian.org.
- **Step two:** install VirtualBox and create a new VM. Use Expert Mode and skip unattended installation; unattended installation often fails.
- **Recommended settings:** Type Linux, Version Debian (64-bit), Memory 4000 MB (2000 MB can work but is slow), disk 60 GB dynamically allocated.
- **Step three:** attach the ISO as a virtual CD-ROM, start the VM, and install.

My own idea/question: The Skip Unattended Install tip is easy to miss and clearly comes from real experience. It shows the guide was written by someone who actually hit the problem, not just copied documentation.

Source: Heinonen 2026 / Karvinen, Install Debian on Virtualbox (updated 2024): https://terokarvinen.com/2021/install-debian-on-virtualbox/

Karvinen 2020: Command Line Basics Revisited

- The command line is older than the internet, and it is still the most efficient way to automate tasks.
- $ is the normal user prompt; you do not type it.
- Navigation: pwd prints the working directory, ls lists files, cd changes directory, cd .. goes up one level.
- Viewing files: less reads a file page by page space for next page, b for previous, q to quit.
- File operations: nano edits text, mkdir creates folders, mv moves or renames, cp -r copies recursively, rm and rmdir delete.
- SSH: ssh user@example.com opens a remote shell; scp -r FOLDER user@example.com:public_html/ copies a folder to a remote server.

My own idea/question: The warning that there is no trash with rm; think before you type is the command-line equivalent of a police caution. Deleting the wrong file destroys evidence. Same discipline applies here.

Source: Karvinen 2020, Command Line Basics Revisited: https://terokarvinen.com/2020/command-line-basics-revisited/

Karvinen 2022: Cracking Passwords with Hashcat

- Systems store hashes, not plaintext passwords. Hashing is one-way; you cannot reverse it mathematically.
- The example hash 6b1628b016dff46e6fa35684be6acc96 is used throughout. The answer is summer.

- Install Hashcat and get a wordlist. Rockyou is the most popular dictionary, with over 14 million words.
- Identify the hash type: hashid or hashcat --example-hashes shows the -m number. This hash is MD5 (mode 0).
- Crack: hashcat -m 0 6b1628b016dff46e6fa35684be6acc96 /usr/share/wordlists/rockyou.txt
- Output: Status: Cracked, Hash.Type: MD5, password: summer.

My own idea/question: The article is honest about the legal boundary: Use of penetration testing techniques requires legal and ethical considerations. In my old work, the difference between a lawful search and an unlawful one was often a single authorization. The same line exists here, and it is the line between a lab exercise and a criminal offence.

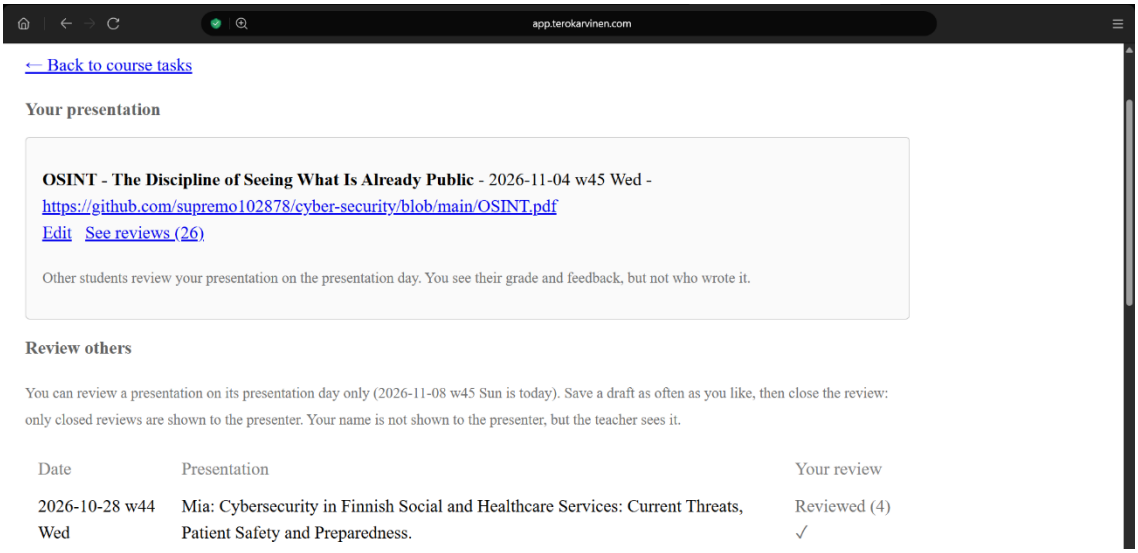
Source: Karvinen 2022, Cracking Passwords with Hashcat: https://terokarvinen.com/2022/cracking-passwords-with-hashcat/

a) Reserve your presentation title in Laksu

I reserved my presentation title in Laksu for my assigned date. The title is:

OSINT – The Discipline of Seeing What Is Already Public

The topic fits the weekly theme of starting a cyber lab: OSINT is the first reconnaissance step before any technical lab work, and it is legal, passive, and teachable without a programming background. It also uses my police investigation background as a strength.



b) Linux – Install Debian 13 "Trixie" on a virtual machine

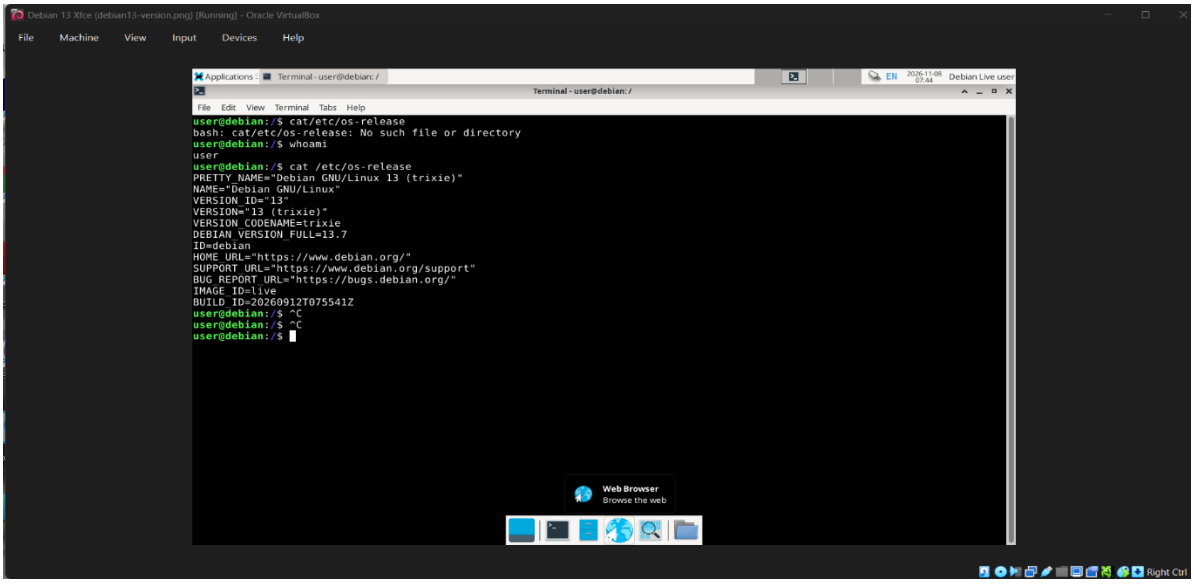
What I did

1. Downloaded the ISO. From debian.org, under Try Debian live, I chose Live Xfce (amd64) before installing.

2. Confirmed the live session version. Booted the ISO in VirtualBox, opened a terminal, and ran:

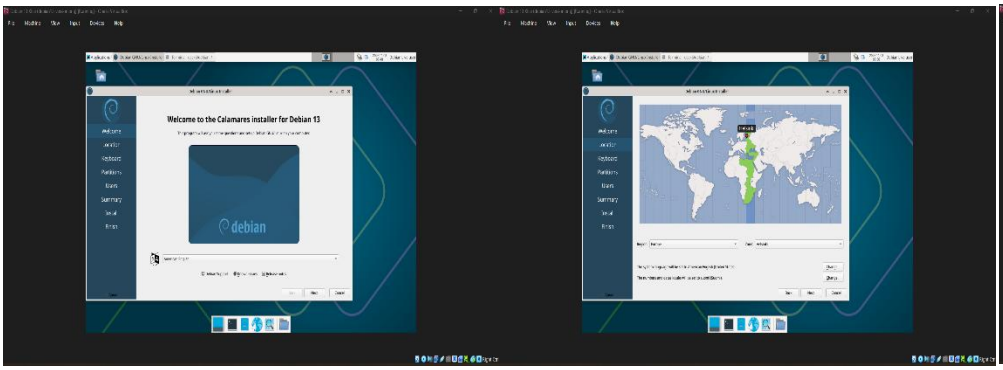
cat /etc/os-release

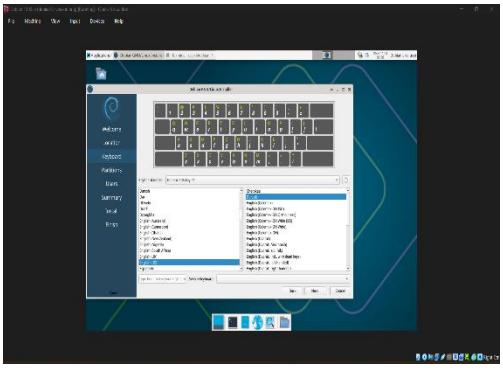
Output included VERSION=13 (trixie).

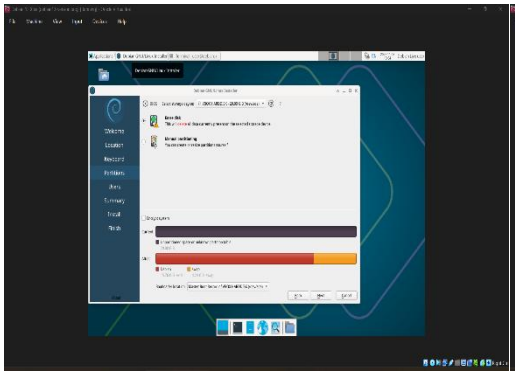


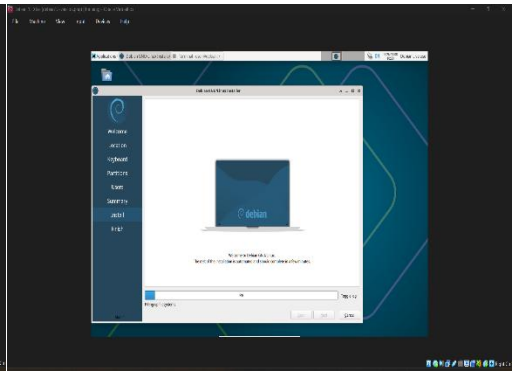
3. Created the VM. In VirtualBox: name Debian 13 Xfce, Type Linux, Version Debian (64-bit), Memory 4096 MB, disk 20 GB dynamically allocated. Skipped unattended installation.

4. Installed Debian. From the live session: language English, location Finland, keyboard English (US). Partitioning: Erase disk (virtual disk only — Windows untouched). User watchtower, computer name watchtower-lab. Password written down on paper. Installation completed and rebooted into the installed system.

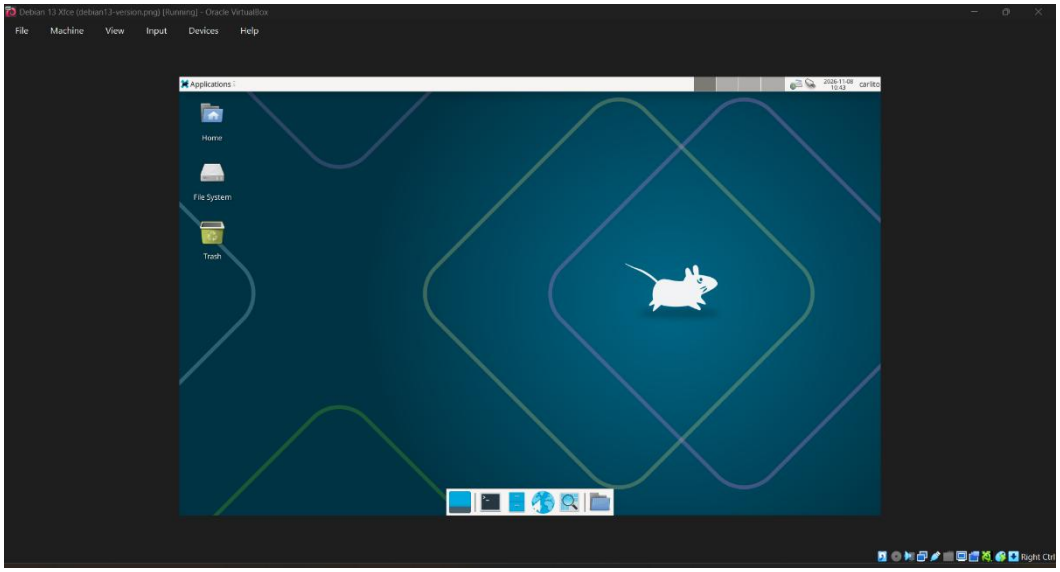


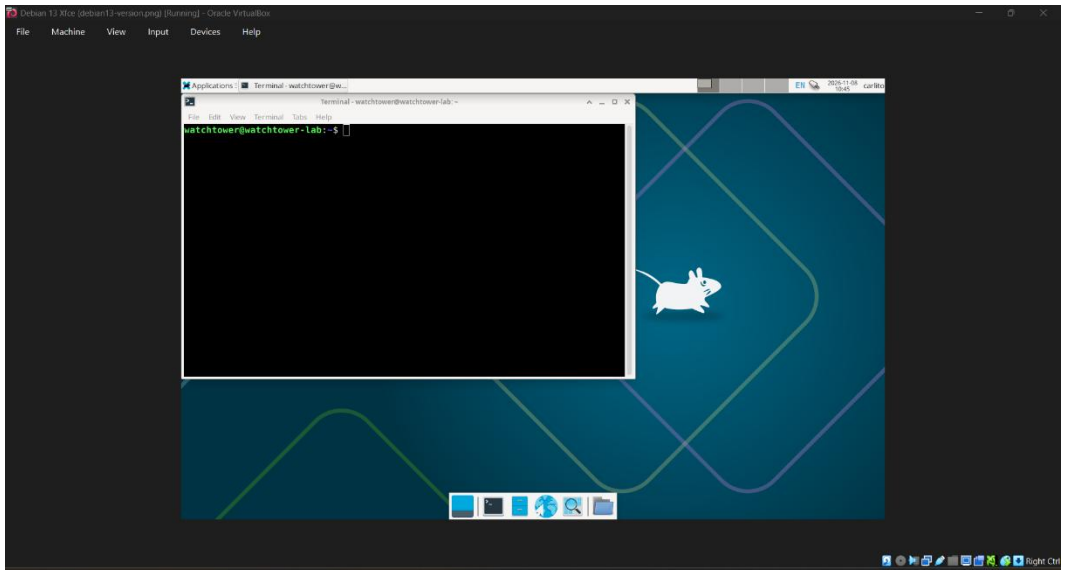


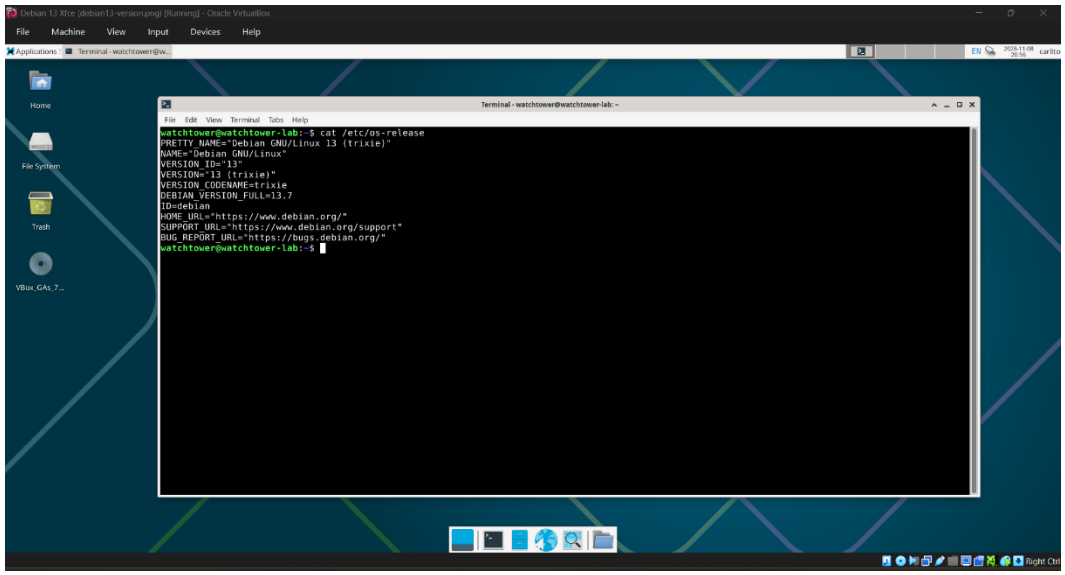




5. Logged in and confirmed the installed system version. Opened a terminal and ran cat /etc/os-release — again `VERSION="13 (trixie)"`.



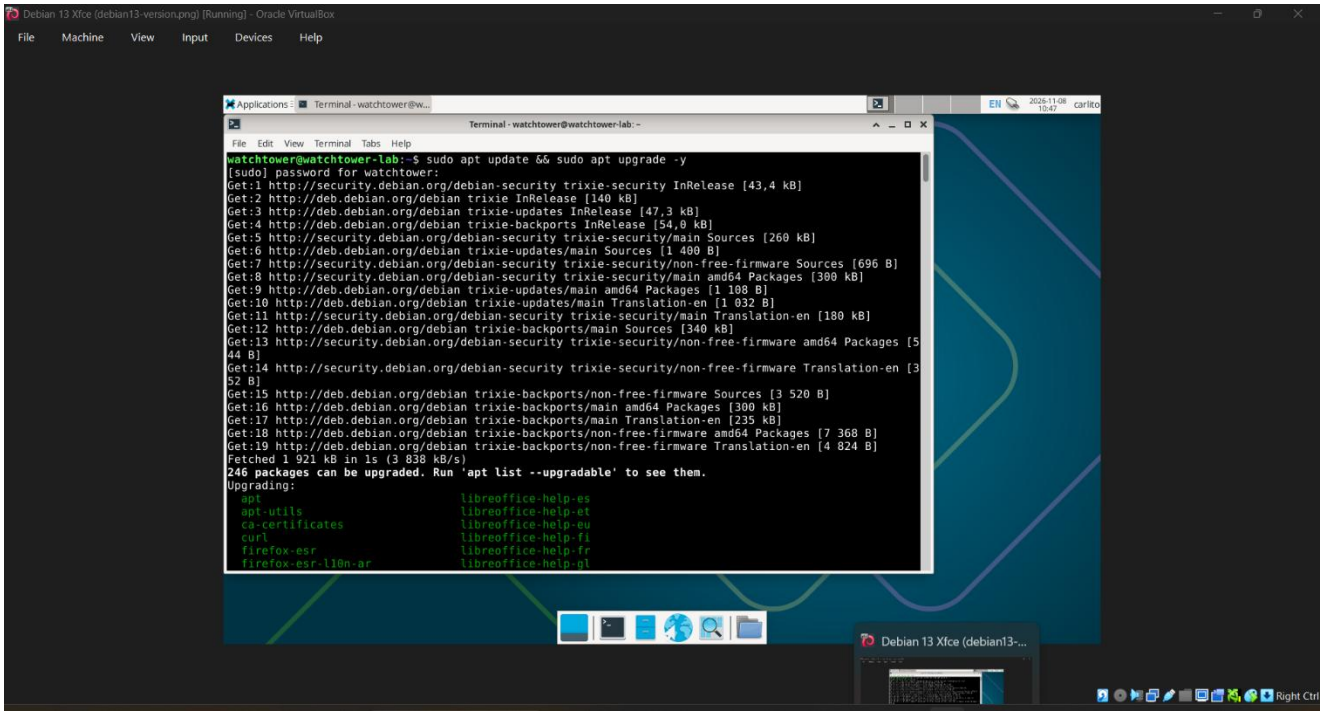


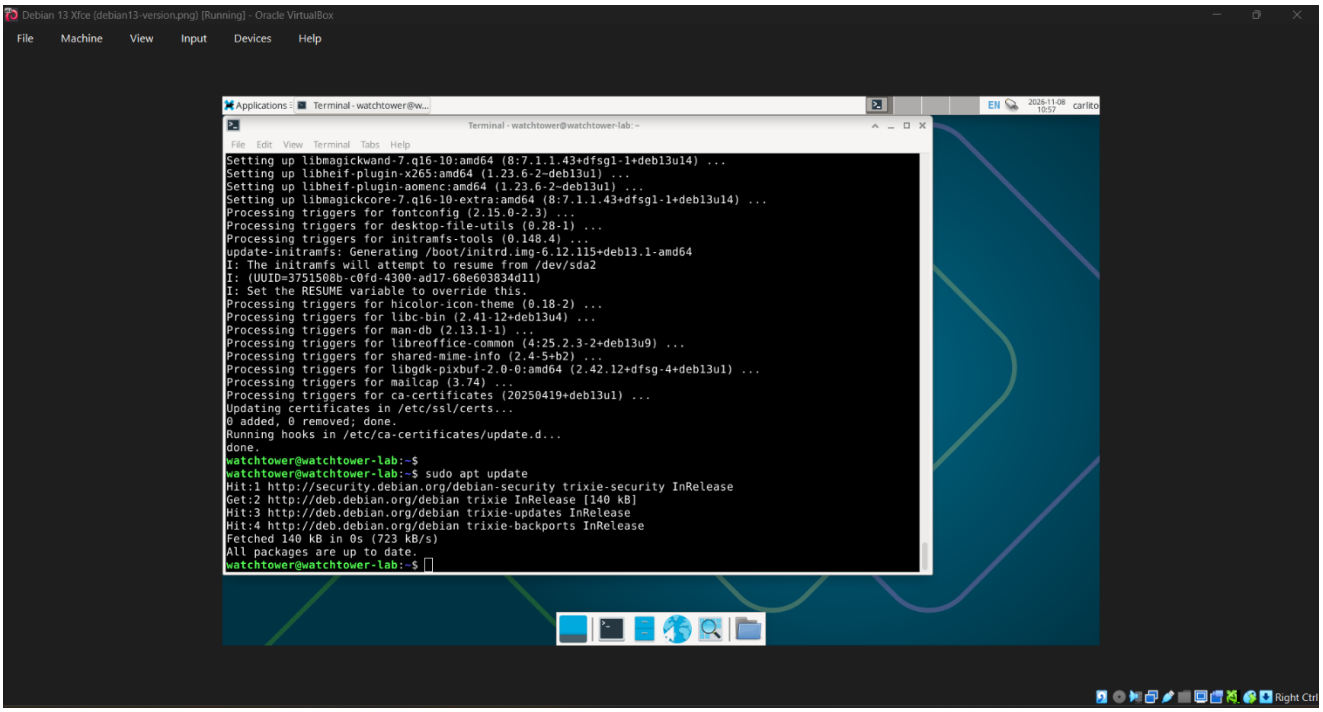


6. Updated the system:

sudo apt update && sudo apt upgrade -y

After it completed, ran sudo apt update again to confirm, which ended with All packages are up to date.





7. Took a VirtualBox snapshot of the clean system (Machine → Take Snapshot).

Problems and fixes

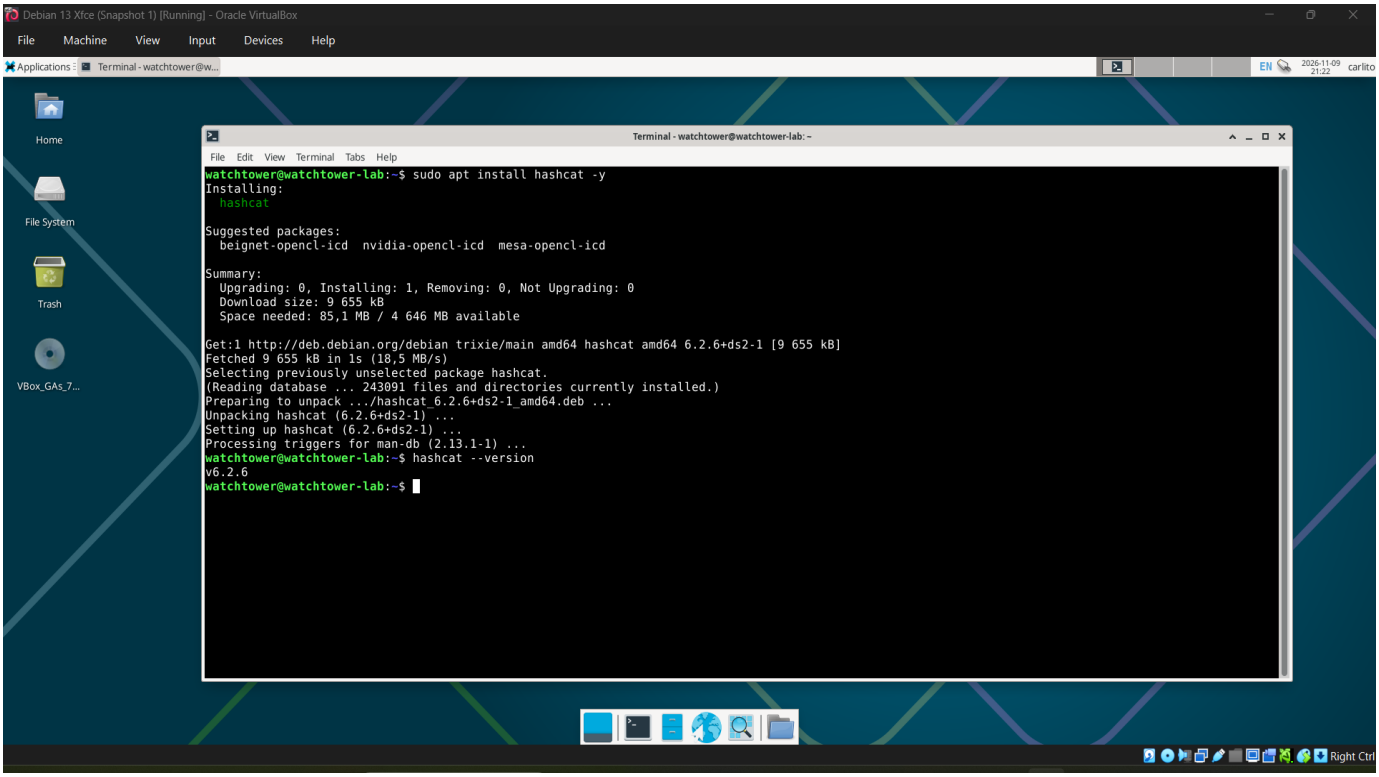
Problem	Fix
Left the live session idle it locked and asked for login.	Debian Live default credentials: username user, password live. Logged back in and continued.
After installation, the VM rebooted into the live session again.	Removed the ISO from the virtual optical drive: Devices → Optical Drives → Remove disk from virtual drive. Then rebooted.
VM was created with only ~1.5 GB RAM.	Changed Settings → System → Motherboard → Base Memory to 4096 MB. This later fixed the Hashcat memory error.

c) Crack the hash 6b1628b016dff46e6fa35684be6acc96

What I did

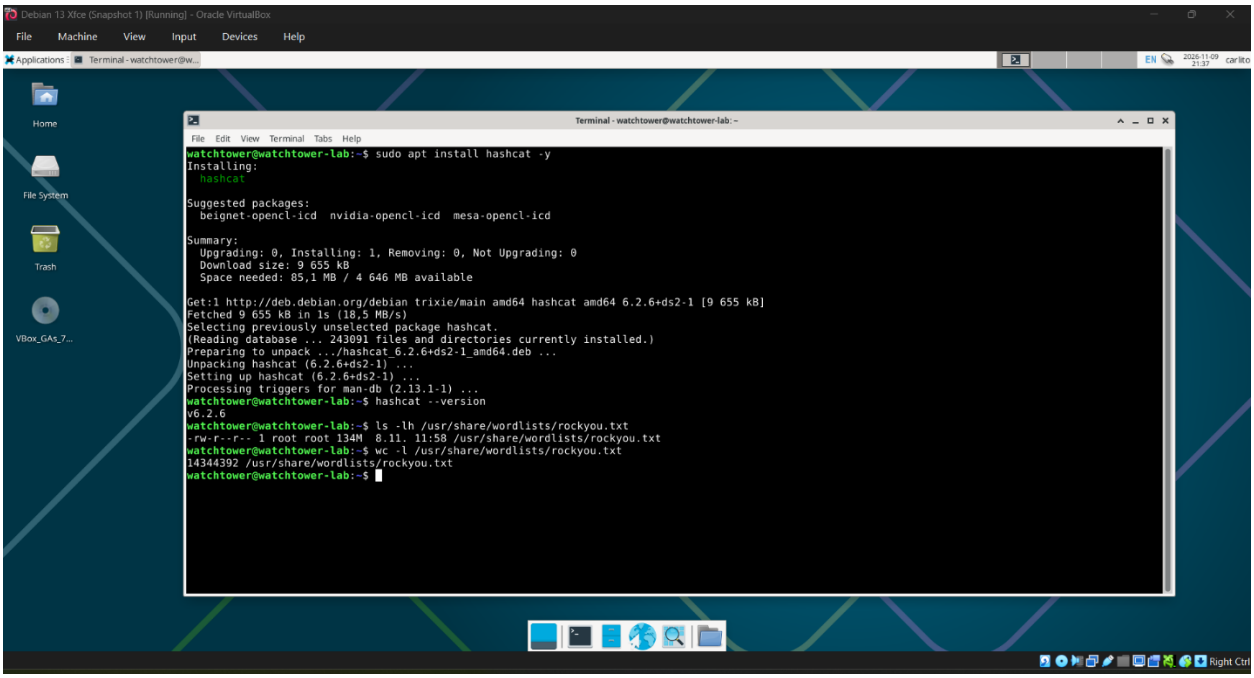
Step 1: Installed Hashcat on the Debian VM.

I installed Hashcat from the Debian repositories with ***sudo apt install hashcat -y***. To confirm the installation was complete and the tool was available on the command line, I checked the version with ***hashcat --version***, which returned v6.2.6.



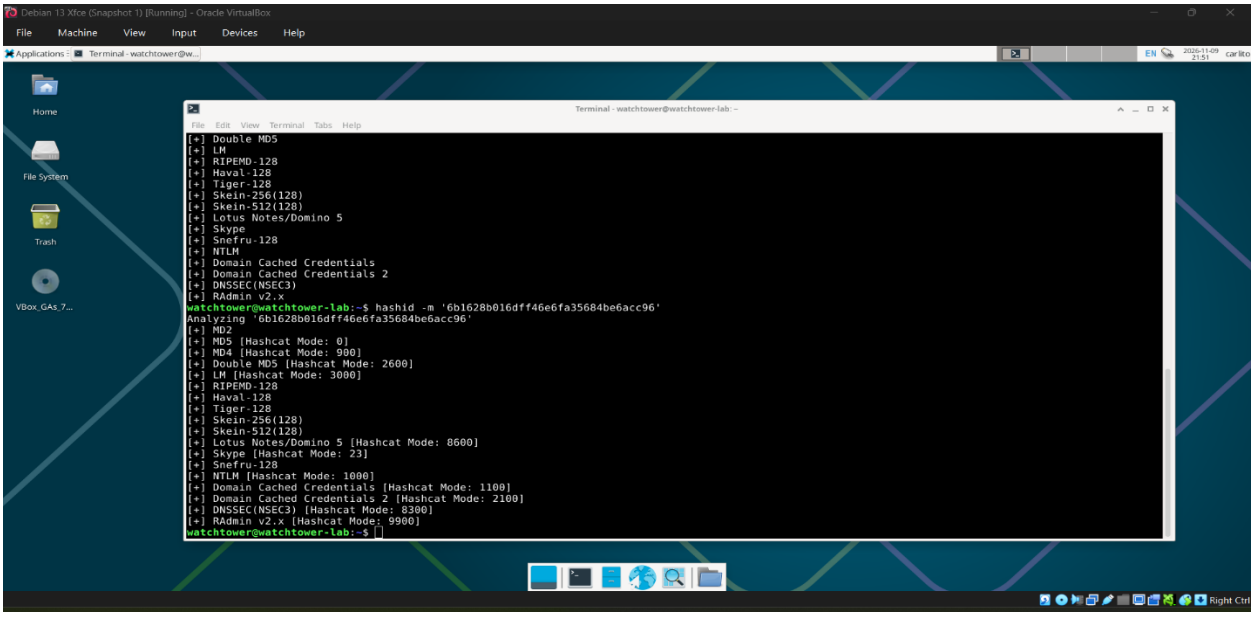
Step 2: Confirmed the full wordlist.

The wordlists and seclists packages aren't in the standard Debian repositories, and wget wasn't installed, so I used curl (already present) to download rockyou from Kali's GitLab repository in an earlier session. Before cracking, I verified the uncompressed file was the real, full list with ***ls -lh /usr/share/wordlists/rockyou.txt*** and ***wc -l /usr/share/wordlists/rockyou.txt***. The output showed a 134 MB file with 14,344,392 lines, confirming the full rockyou dictionary was in place.



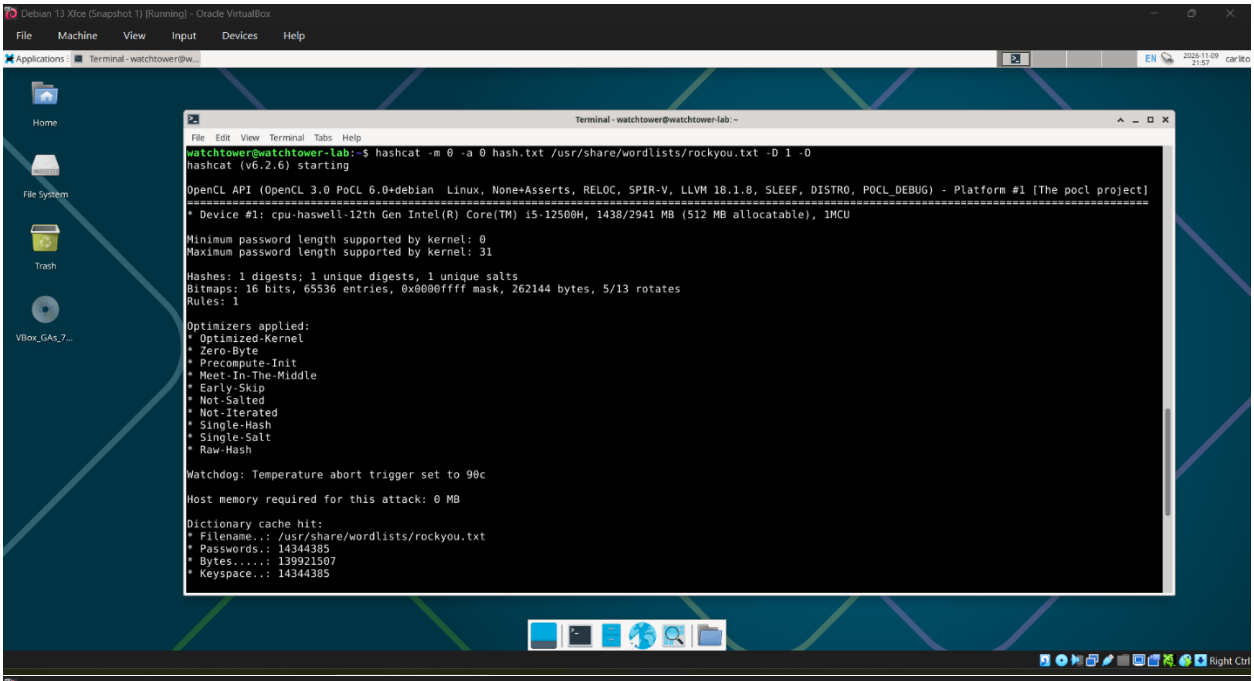
Step 2b: Identified the hash type.

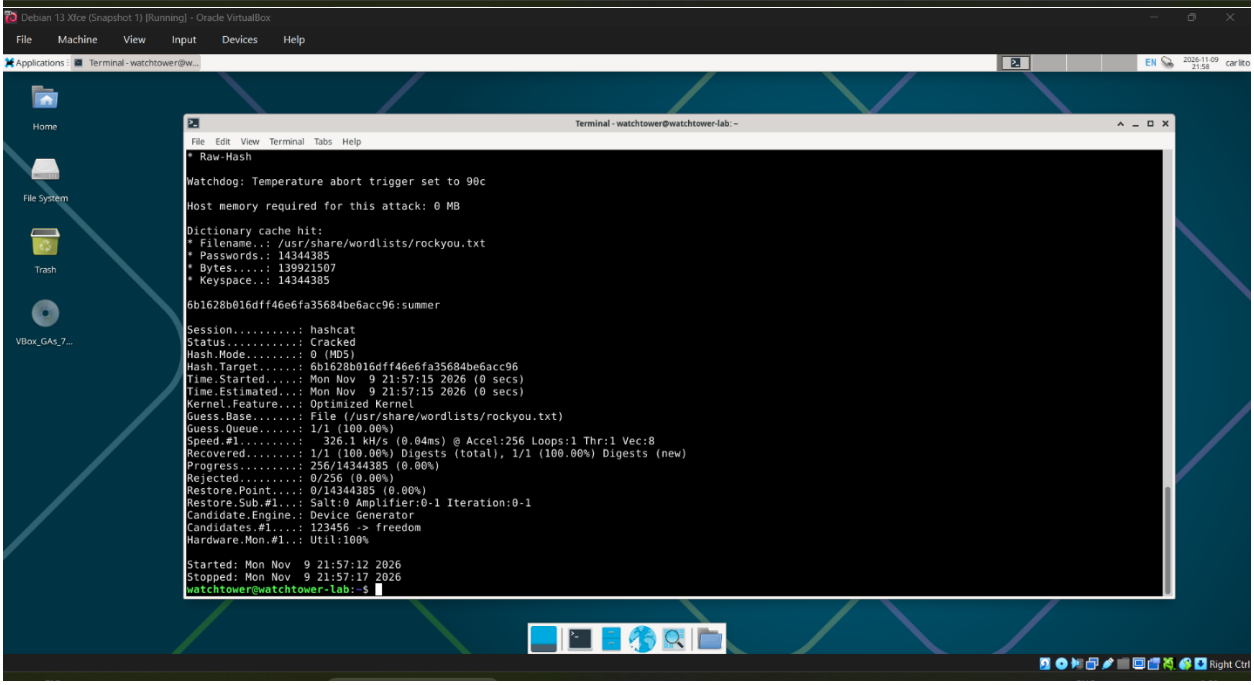
The Karvinen 2022 article's workflow includes identifying the hash before cracking it, because the hash type determines the -m mode number Hashcat needs. I ran ***hashid -m '6b1628b016dff46e6fa35684be6acc96'***, which listed the possible hash types along with their Hashcat mode numbers. The relevant line was [+] MD5 [Hashcat Mode: 0], confirming that mode 0 (MD5) was the correct choice for the next step. These matches both the 32-character hexadecimal format of the hash and the hash type stated in the article.



Step 3: Ran Hashcat.

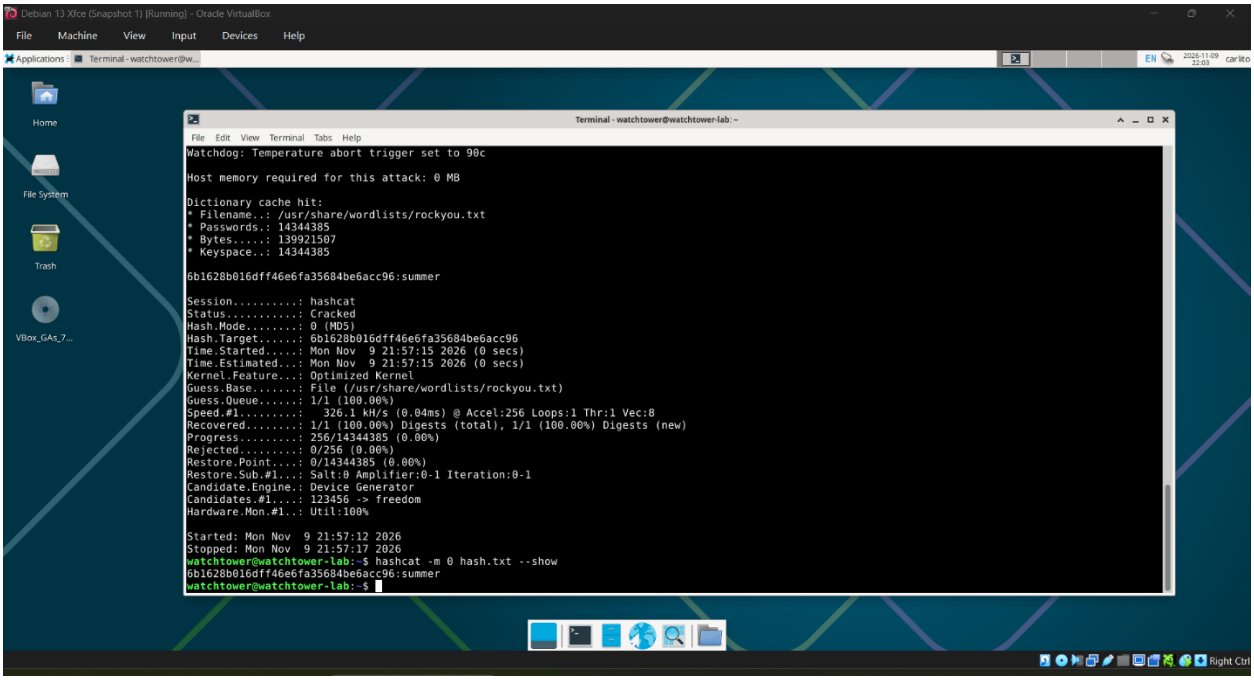
With the hash type confirmed as MD5, I ran Hashcat in dictionary-attack mode against the full rockyou wordlist. The command used -m 0 for MD5, -a 0 for a dictionary attack, -D 1 to force CPU mode (the VM has no GPU passthrough), and -O to use the optimized kernel, which was necessary because of the VM's memory limits. The crack finished in about one second and reported Status.....: Cracked with the recovered plaintext summer, matching the answer stated in the Karvinen 2022 article.





Step 4: Confirmed the cracked password with --show.

To display only the cracked result the hash and its plaintext password without the full run log, I ran ***hashcat -m 0 hash.txt --show***. It printed 6b1628b016dff46e6fa35684be6acc96:summer, confirming that the MD5 hash 6b1628b016dff46e6fa35684be6acc96 corresponds to the password summer, exactly as stated in the Karvinen 2022 article.



References

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Kali Linux. 2019. wordlists package – contains the rockyou.txt wordlist (134 MB). GitLab. Available at: https://gitlab.com/kalilinux/packages/wordlists. Accessed: 8.11.2026. [reference:3]

Hashcat. 2026. Hashcat Advanced Password Recovery – hash mode 0 (MD5). Available at: https://hashcat.net/hashcat/. Accessed: 8.11.2026. [reference:4]

Hashid. 2026. Hashid – identify hash types and suggest hashcat mode. Available at: https://github.com/psypanda/hashID. Accessed: 8.11.2026. [reference:5]

AI disclosure: I used an AI assistant to help structure the report, troubleshoot VirtualBox and OpenCL memory errors, and locate a working rockyou download URL. All commands, outputs, and screenshots are from my own computer. The crack result (`summer`) was verified against the Karvinen 2022 article's stated answer.